

Complete token-level view for the Figure 1 disagreement case | Part 1/5

Tokens 1-1,842 | 8,619 response tokens | GeoMean s=1.0025 <= 1.005 (accept) | CARM s=1.0248 > 1.02 (reject)

■ $\log r_t > 0$ ■ $\log r_t < 0$

tokens 1-895

Here 's a thinking process to solve the problem : \n\n

1 . ** Under stand the Goal : ** The user wants to find the value of f

(- 1 0 0)\$ given a functional equation and a specific definition of f

(x)\$ on an interval . Finally , express the result as a fraction m/n

\$ and find $m+n$ \$. \n\n

2 . ** An alyze the Given Information : ** \n

* Domain : \$ $f(x)$ is defined on $\mathbb{R}$. \n

* Functional Equation : \$ $f(x+3) \cdot f(x-4) = -1$ \$ for

all $x \in \mathbb{R}$. \n

* Definition on an interval : \$ $f(x) = \log_2(9-x)$ \$ for $0 \leq x < 7$. \n

* Target : Calculate \$ $f(-100)$ \$. \n

* Output format : \$ $m+n$ \$ where the answer is $\frac{m}{n}$, $n > 0$, $\gcd(m, n) = 1$ \$, though the prompt just says format \$ m/n \$ and asks for \$ $m+n$ \$. \n\n

3 . ** Explore the Functional Equation : ** \n

Let the equation be \$ $f(x+3) \cdot f(x-4) = -1$. \n

We want to find relations between values of \$ f \$ at different points . \n

Let 's replace \$ x \$ with \$ $x+4$ \$ to see a pattern in the arguments . \n

\$ $f((x+4)+3) \cdot f((x+4)-4) = -1$ \implies \$ $f(x+7) \cdot f(x) = -1$. \n

This looks promising because it links \$ $f(x+7)$ \$ and \$ $f(x)$ \$. \n

Let 's check if this relation holds for all \$ x \$. Yes , it 's derived from the original identity for all \$ x \$. \n

So , \$ $f(x+7) = -\frac{1}{f(x)}$. \n\n

Let 's look at the next step to find a periodic ity . \n

Apply the relation again for \$ $f(x+7)$: \n

\$ $f(x+14) = -\frac{1}{f(x+7)} = \frac{1}{f(x)}$. \n

So , \$ $f(x+14) = f(x)$ \$ for all \$ x \$. \n

The function \$ $f(x)$ \$ is periodic with period \$ $T = 14$. \n\n

4 . ** Calculate Target Argument : ** \n

We need to find \$ $f(-100)$ \$. \n

Since \$ $f(x)$ \$ has period 14 , we can shift the argument by multiples of 14 to bring it into the known interval \$ $[0, 7)$ \$. \n

We want to find an integer \$ k \$ such that \$ $-100 + 14k \in [0, 7)$. \n

Let 's approximate : \$ $-100 / 14 \approx -7.14$. \n

So \$ k \$ could be 7 , 8 , or 8 . \n

Let 's try \$ $k = 8$ \$: \n

\$ $-100 + 14(8) = -100 + 112 = 12$. \n

Wait , the definition \$ $f(x) = \log_2(9-x)$ \$ is for \$ $0 \leq x < 7$. \n

\$ $x = 12$ \$ is not in \$ $[0, 7)$ \$. \n

However , we know \$ $f(12) = f(12-14) = f(-2)$ \$. \n

Let 's try to map \$ -100 \$ to the range \$ $[0, 7)$ \$. \n

\$ $-100 + 14k \in [0, 7)$. \n

Let 's calculate \$ $-100 \bmod 14$ \$. \n

\$ $-100 = 7 \cdot (-14) + 2$ \$. So \$ $-100 \equiv 2 \pmod{14}$ \$. \n

tokens 896-1,842

Thus \$ $-100 \equiv 2 \pmod{14}$ \$. \n

This means \$ $f(-100) = f(2)$ \$. \n

Also \$ $f(12) = f(12-14) = f(-2)$ \$. \n

But we need the value in terms of the definition \$ $f(x) = \log_2(9-x)$ \$. \n

(9 -x)\$ which is defined on \$ $[0, 7)$ \$. \n

So we need to evaluate \$ $f(-100)$ \$ by relating it to a value \$ $x \in [0, 7)$ \$. \n\n

Let 's check the relation \$ $f(x+7) = -1/f(x)$ \$ again . \n

If we have a value in \$ $[0, 7)$ \$, say \$ $x \in [0, 7)$ \$, then \$ $f(x+7) = -1/f(x)$ \$. Note that \$ $x+7$ \$ would be in \$ $[7, 14)$ \$. \n

We have \$ $f(-100)$ \$. \n

We found \$ $-100 \equiv 2 \pmod{14}$ \$. \n

So \$ $f(-100) = f(2)$ \$. \n

We also know \$ $f(x+14) = f(x)$ \$. \n

Wait , I 'm saying \$ $f(-100) = f(2)$ \$. \n

But \$ $2 \notin [0, 7)$ \$. \n

Can we express \$ 2 \$ using the shift relation backwards ? \n

\$ $f(12) = f(12-14) = f(-2)$? No , that goes back to negative . \n

Let 's look at the relation \$ $f(x+7) \cdot f(x) = -1$. \n

This implies \$ $f(x+7) = -1/f(x)$. \n

We want to use the definition \$ $f(x) = \log_2(9-x)$ \$ for \$ $0 \leq x < 7$. \n

The value 12 is outside this range . \n

However , we know the functional equation holds for all real \$ x \$. \n

So we can use \$ $f(x+7) \cdot f(x) = -1$ \$ to relate a value at \$ 12 \$ to a value at \$ 5 \$ (since \$ $12-7 = 5$ \$) . \n

Let 's check : \$ $5 \in [0, 7)$ \$. \n

So , \$ $f(12) \cdot f(5) = -1$ \implies \$ $f(12) = -1/f(5)$. \n

Since \$ $f(-100) = f(12)$ \$, then \$ $f(-100) = -1/f(5)$ \$. \n\n

Now we need to calculate \$ $f(5)$ \$. \n

Since \$ $0 \leq 5 < 7$ \$, we can use the formula directly . \n

\$ $f(5) = \log_2(9-5) = \log_2(4) = 2$. \n\n

Therefore , \$ $f(-100) = -1/2 = -\frac{1}{2}$ \$. \n\n

5 . ** Double Check the Logic : ** \n

* Functional Equation : \$ $f(x+3) \cdot f(x-4) = -1$ \$. \n

* Let \$ $x = y+4$ \$. Then \$ $f(y+7) \cdot f(y) = -1$ \$. \n

* Im plies \$ $f(y) = -1/f(y+7)$ \$. \n

* Period icity : \n

\$ $f(y+14) = f(y+7+7) = -1/f(y+7) = -1/(-1/f(y)) = f(y)$ \$. \n

Period is 14 . \n

* Target : \$ $f(-100)$ \$. \n

* \$ $-100 = -7 \cdot 14 + 2 = -98 + 2$ \$. No . \n

\$ $-100 / 14 = -7.1428 \dots$ \$ \n

\$ $-7 \cdot 14 = -98$ \$. So \$ $-100 = -7 \cdot 14 + 2$ \$. \n

2 \$. \n

Thus \$ $-100 \equiv 2 \pmod{14}$ \$. \n

So \$ $f(-100) = f(-2)$ \$. \n

Complete token-level view for the Figure 1 disagreement case | Part 2/5

Tokens 1,843-3,632 | Continuation of the same unabridged tokenizer-aligned response.

■ $\log r_t > 0$ ■ $\log r_t < 0$

tokens 1,843-2,778

* Now we have $f(-2)$. Is -2 in $[0, 7]$? No. $\backslash n$

* We need to use the relation $f(x+7)f(x) = -1$ to map -2 to the interval $[0, 7]$. $\backslash n$

* Let $x = -2 - 7 = -9$. Then $f(-9-7)f(-9) = f(-16)f(-9)$? No. $\backslash n$

* We want x $\{ \text{new} \}$ in $[0, 7]$ such that x $\{ \text{new} \}$ is related to -2 by a shift of 7 . $\backslash n$

* $-2 + 7 = 5$. $\backslash n$

* So consider $f(5)$. We have the relation $f(5)f(-2) = -1$. $\backslash n$

* So $f(-2) = -1/f(5)$. $\backslash n$

* Check validity: 5 in $[0, 7]$. Formula $f(x) = \backslash \log_{-2}(9-x)$ applies. $\backslash n$

* Calculate $f(5) = \backslash \log_{-2}(9-5) = \backslash \log_{-2}(4) = 2$. $\backslash n$

* So $f(-2) = -1/2$. $\backslash n$

* Since f is periodic with period 14 , $f(-100) = f(-2) = -1/2$. $\backslash n$

* Wait, is $f(-100) = f(-2)$ consistent with earlier mod calc? $\backslash n$

$-100 = -8 \times 14 + 12$. So $f(-100) = f(12)$. $\backslash n$

$12 - 7 = 5$. $\backslash n$

$f(12)f(5) = -1$ implies $f(12) = -1/2$. $\backslash n$

$-100 \equiv 12 \pmod{14}$. $\backslash n$

$-100 + 112 = 12$. Correct. $\backslash n$

$-100 = -7 \times 14 - 2$. So $-100 \equiv -2 \pmod{14}$. $\backslash n$

$12 \equiv -2 \pmod{14}$. Correct. $\backslash n$

* Also, is there any constraint violation? $\backslash n$

The definition is given for $0 \leq x < 7$. $\backslash n$

We used $f(5)$ which is in that range. $\backslash n$

We used $f(x+7)f(x) = -1$. This holds for all real x . $\backslash n$

So $f(5)f(-2) = -1$. $\backslash n$

$f(-2) = -1/2$. $\backslash n$

$f(-100) = f(-2) = -1/2$. $\backslash n$

* Result is $-1/2$. $\backslash n$

* Format m/n . So $m = -1$, $n = 2$. $\backslash n$

* Usually in these problems m, n are integers, $n > 0$. If reduced, $\text{gcd}(m, n) = 1$. $\backslash n$

* $m = -1$, $n = 2$. $m+n = 1$. $\backslash n$

* Or maybe they expect m to be positive? m/n format. $\backslash n$

Usually implies rational number. Negative numerator is allowed. $\backslash n$

* Let's check if there's ambiguity in the sign. $f(x) = \backslash \log_{-2}(9-x)$. $\backslash n$

* Domain $0 \leq x < 7$. For $x = 5$, $9-5 = 4$. $\backslash \log_{-2} 4 = 2$. Positive. $\backslash n$

* $f(-100) = -1/2$. $\backslash n$

* $m = -1$, $n = 2$. $m+n = 1$. $\backslash n$

* Let's re-read carefully: m/n format. Often implies lowest terms. $\backslash n$

* What if the fraction is simplified differently? No, $-1/2$ is simplest form. $\backslash n$

* Are m, n required to be coprime? Give the value of $m+n$

tokens 2,779-3,632

* If I used $-2/4$, sum is 2 . If $-1/2$, sum is 1 . $\backslash n$

Standard mathematical convention implies coprime integers with $n > 0$. $\backslash n$

I will assume this standard convention. $\backslash n$

Let's re-verify the period derivation. $\backslash n$

$f(x+3)f(x-4) = -1$. $\backslash n$

Replace x with $x+4$: $f(x+7)f(x) = -1$. $\backslash n$

Periodicity: $\backslash n$

$f(x+14) = \backslash \frac{(-1)}{f(x+7)} = \backslash \frac{(-1)}{-1/f(x)} = f(x)$. Correct. $\backslash n$

Modulo arithmetic: $\backslash n$

$-100 = -7 \times 14 - 2 \equiv -2 \pmod{14}$. $\backslash n$

$-100 = -8 \times 14 + 12 \equiv 12 \pmod{14}$. $\backslash n$

$f(-100) = f(-2) = f(12)$. $\backslash n$

Using $f(x+7)f(x) = -1$ for $x = -2$: $\backslash n$

$f(-2+7)f(-2) = -1$ implies $f(5)f(-2) = -1$. $\backslash n$

Using $x = 5$: $\backslash n$

$f(5) = \backslash \log_{-2}(9-5) = \backslash \log_{-2}(4) = 2$. $\backslash n$

$f(-100) = -1/2$. $\backslash n$

Is there any potential trap? $\backslash n$

* defined on $\mathbb{R}$. $0 \leq x < 7$. $\backslash n$

Is it possible $f(x) = 0$ somewhere? $\backslash n$

$\backslash \log_{-2}(9-x)$ is never 0 unless $9-x = 1$ implies $x = 8$. $\backslash n$

f , which is outside $[0, 7]$. $\backslash n$

If $f(x) = 0$, the relation $f(x+7)f(x) = -1$ would imply $0 = -1$, contradiction. $\backslash n$

So $f(x) \neq 0$. $\backslash n$

Is the domain of log valid? $9-x > 0$ implies $x < 9$. $\backslash n$

Interval $[0, 7]$ satisfies this. $\backslash n$

So the function is well-defined. $\backslash n$

Is there another shift? $\backslash n$

$f(x+3)f(x-4) = -1$. $\backslash n$

Maybe use this directly? $\backslash n$

Let $x-4 = -100$ implies $x = -96$. $\backslash n$

Then $f(-96+3)f(-100) = -1$ implies $f(-93)f(-100) = -1$. $\backslash n$

Need $f(-93)$. $\backslash n$

$-93 = -7 \times 14 - 5 \equiv -5 \pmod{14}$. $\backslash n$

$-93 \equiv 9 \pmod{14}$. $\backslash n$

$f(-93) = f(9)$. $\backslash n$

$9 = 2 + 7$. $\backslash n$

$f(9)f(2) = -1$ implies $f(9) = -1/f(2)$. $\backslash n$

2 in $[0, 7]$. $f(2) = \backslash \log_{-2}(9-2) = \backslash \log_{-2}(7)$. $\backslash n$

This looks messy. $f(-93)$ needs to be calculated. $\backslash n$

Let's stick to the period derivation. $\backslash n$

$f(x+14) = f(x)$. $\backslash n$

We established $f(-100) = f(12)$. $\backslash n$

Then $f(12)f(5) = -1$ (since $12-7=5$). $\backslash n$

$f(5)$ is easily calculated. $\backslash n$

$f(-100) = -1/f(5)$. $\backslash n$

Complete token-level view for the Figure 1 disagreement case | Part 3/5

Tokens 3,633-5,377 | Continuation of the same unabridged tokenizer-aligned response.

log $r_t > 0$ log $r_t < 0$

tokens 3,633-4,521

\$ f (5) = \log _{-2} (4) = 2 \$. \n

\$ f (- 1 0 0) = - 1 / 2 \$. \n\n

Let 's double check \$ f (x + 7) \$ if \$ (x) = - 1 \$ \$. \n

Original : \$ f (x + 3) \$ if \$ (x - 4) = - 1 \$ \$. \n

Shift \$ x \setminus \$ to \$ x + 7 \$ \$. \n

\$ f (x + 7 + 3) \$ if \$ (x + 7 - 4) = f (x + 1 0) \$ if \$ (x + 3) = - 1 \$ \$. \n

\n

This doesn 't match \$ f (x + 7) \$ if \$ (x) \$ \$. \n

Wait , derivation was : \n

Let \$ y = x - 4 \$ \$. Then \$ x = y + 4 \$ \$. \n

\$ f (y + 4 + 3) \$ if \$ (y) = - 1 \setminus \$ impl ies \$ f (y + 7) \$ if \$ (y) = - 1 \$ \$. \n

\n

This derivation is correct . \n

Wait , earlier I wrote " Replace \$ x \$ with \$ x + 4 \$ " \$. \n

Original equation : for any \$ x \$ \$, \$ f (x + 3) \$ if \$ (x - 4) = - 1 \$ \$. \n

Let 's substitute specific \$ x \$ \$. \n

If we want to link \$ f (x) \$ and \$ f (x + 7) \$, we need the arguments

of \$ f \$ in the equation to match \$ x \$ and \$ x + 7 \$ \$. \n

In \$ f (x + 3) \$ if \$ (x - 4) \$, the difference between arguments is \$ (x + 3) - (x - 4) = 7 \$ \$. \n

So yes , \$ f (u) \$ if \$ (v) \$ with \$ u - v = 7 \$ equals \$ - 1 \$ \$. \n

Let \$ u = x + 7 \$ \$, \$ v = x \$ \$ u - v = 7 \$ \$. \n

Then \$ f (u) \$ if \$ (v) = - 1 \$ \$, so \$ f (x + 7) \$ if \$ (x) = - 1 \$ \$. \n

Yes , derivation stands . \n\n

Now , check the mapping from \$ - 1 0 0 \$ \$ to \$ (0 , 7) \$ \$. \n

\$ - 1 0 0 \setminus \$ equiv \$ 1 2 \setminus \text{p mod } \{ 1 4 \} \$ \$. \n

\$ 1 2 \setminus \$ equiv \$ 5 \setminus \text{p mod } 7 \$ \$. Wait . \n

\$ f (x + 7) = - 1 / f (x) \$ \$. \n

\$ f (x + 1 4) = f (x) \$ \$. \n

So we map \$ - 1 0 0 \$ \$ by adding multiples of \$ 1 4 \$ \$. \n

\$ - 1 0 0 + 1 1 2 = 1 2 \$ \$. \n

We want an input to the function definition \$ f (t) = \log _{-2} (9 - t) \$ for \$ t \setminus \$ in \$ [0 , 7) \$ \$. \n

We have \$ f (1 2) \$ \$. \n

\$ f (1 2) \$ if \$ (5) = - 1 \$ because \$ 1 2 = 5 + 7 \$ \$. \n

Is \$ 5 \setminus \$ in \$ [0 , 7) \$? Yes . \n

So \$ f (5) = \log _{-2} (9 - 5) = 2 \$ \$. \n

\$ f (1 2) = - 1 / 2 \$ \$. \n

\$ f (- 1 0 0) = f (1 2) = - 1 / 2 \$ \$. \n

Seems solid . \n\n

Is there any ambiguity about \$ m , n \$ being integers ? \n

Usually , questions like " find \$ m + n \$ where \$ m / n \$ is the value " imply

integer numerator and denominator in lowest terms . \n

\$ m = - 1 , n = 2 \$ \$. \n

\$ m + n = 1 \$ \$. \n

Sometimes they want positive \$ n \$ \$. Yes , usually \$ n > 0 \$ \$. \n

Sometimes they define \$ m , n \$ as positive integers and add a sign to

the value ? No , \$ m / n \$ format usually allows negative \$ m \$ \$. \n

If the answer was \$ - 1 / 2 \$, then \$ m = - 1 , n = 2 \$ \$. \n

If the problem implied \$ m , n \setminus \$ in \$ \setminus \text{math bb } \{ \mathbb{Z} \} ^{+} \$ and \$ m \$ is

the numerator of the simplified fraction (including sign) , it might be

specified . But standard interpretation is \$ m , n \setminus \$ in \$ \setminus \text{math bb } \{ \mathbb{Z} \} ,

\$ n \setminus \text{ne } 0 , \setminus \text{gcd } (| m | , | n |) = 1 , n > 0 \$ \$. \n

So \$ m = - 1 , n = 2 \$ \$. \n

tokens 4,522-5,377

Sum = 1 . \n\n

Wait , could the log be \$ \setminus \$ in \$? \$ The symbol is \$ \setminus \log _{-2} \$ \$. \n

Could the base be something else ? \$ \setminus \log _{-2} \$ \$. \n

\$ f (5) = \log _{-2} (4) = 2 \$ \$. \n

Calculation is correct . \n\n

Let me check if I missed any subtle constraints . \n

" defined on \$ \setminus \text{math bb } \{ \mathbb{R} \} ^{+} \$ " \$. \n

" for any real number \$ x \$ " \$. \n

Is the periodic ity unique ? \n

\$ f (x + 1 4) = f (x) \$ \$. Could it have a smaller period ? \n

\$ f (x + 7) = - 1 / f (x) \$ \$. \n

Since \$ f (x) \$ is not ident ically \$ 1 \$ or \$ - 1 \$ (value is 2

), period cannot be 7 (unless \$ f (x + 7) = f (x) \setminus \$ impl ies \$ f (x)

$= - 1 / f (x) \setminus $ impl ies $ f (x)^2 = - 1 $, impossible for real numbers) .$

\n

So period is exactly \$ 1 4 \$ (or a multiple if \$ f \$ has other

properties not given , but \$ 1 4 \$ is sufficient to solve) . \n

Actually , \$ f (x + 7) = - 1 / f (x) \$ means \$ f (x) \$ is anti

-period ic with period 7 . \n

So \$ f (x + 1 4) = f (x) \$ \$. \n

\$ - 1 0 0 = - 7 \setminus \$ times \$ 1 4 = - 2 \setminus \$ times \$ 7 \$ \$. \n

So \$ f (- 1 0 0) = f (- 2) \$ \$. \n

\$ f (- 2) = - 1 / f (5) \$ \$. \n

Calculation : \$ f (5) = 2 \$ \$. \n

Result \$ - 0 . 5 \$ \$. \n\n

Let 's quickly check \$ x \$ mapping again . \n

\$ f (x) \$ is given for \$ 0 \setminus \text{le } x < 7 \$ \$. \n

We need \$ f (- 1 0 0) \$ \$. \n

\$ - 1 0 0 \setminus \text{p mod } \{ 1 4 \} \$ \$. \n

\$ 1 0 0 = 7 \times 1 4 + 3 0 = 7 \times 1 4 + 2 \times 7 + 2 \times 7 + 1 0 0 \setminus \text{equiv}

\$ 2 \setminus \text{p mod } \{ 1 4 \} \$ \$. \n

\$ - 1 0 0 \setminus \text{equiv } - 2 \setminus \text{equiv } 1 2 \setminus \text{p mod } \{ 1 4 \} \$ \$. \n

So \$ f (- 1 0 0) = f (1 2) \$ \$. \n

\$ 1 2 = 7 + 5 \$ \$. \n

\$ f (1 2) = f (5 + 7) \$ \$. \n

\$ f (x + 7) = - 1 / f (x) \$ \$. \n

\$ f (5 + 7) = - 1 / f (5) \$ \$. \n

\$ 5 \setminus \$ in \$ [0 , 7) \$ \$. \n

\$ f (5) = \log _{-2} (9 - 5) = \log _{-2} (4) = 2 \$ \$. \n

\$ f (1 2) = - 1 / 2 \$ \$. \n

Everything is consistent . \n\n

What if I evaluate \$ f (- 2) \$ directly from the relation \$ f (x + 3) \$ if

\$ (x - 4) = - 1 \$? \n

\$ f (- 2 + 3) \$ if \$ (- 2 - 4) = - 1 \setminus \$ impl ies \$ f (1) \$ if \$ (- 6) = - 1

\$. \n

\$ f (- 6) \$? \$ - 6 \setminus \text{equiv } 8 \setminus \text{p mod } \{ 1 4 \} \$ \$. \$ \$ f (- 6) = f

\$ (8) \$ \$. \n

\$ 8 = - 1 + 7 \$ \$. \$ f (8) = - 1 / f (1) \$ \$. \n

So \$ f (1) (- 1 / f (1)) = - 1 \setminus \$ impl ies \$ - 1 = - 1 \$ \$. T aut

ology . Doesn 't help find value of \$ f (1) \$ \$. \n

We need the value in \$ [0 , 7) \$ \$. \n

\$ f (- 2) \$ \$. \n

\$ f (- 2 + 7) \$ if \$ (- 2) = - 1 \setminus \$ impl ies \$ f (5) \$ if \$ (- 2) = - 1 \$ \$. \n

\$ 5 \setminus \$ in \$ [0 , 7) \$ \$. \$ f (5) = 2 \$ \$. \n

Complete token-level view for the Figure 1 disagreement case | Part 4/5

Tokens 5,378-6,954 | Continuation of the same unabridged tokenizer-aligned response.

log $r_t > 0$ log $r_t < 0$

tokens 5,378-6,188

$f(-2) = -1 \implies f(-2) = -1/2$.
Matches.
Are there any constraints on m, n ?
Usually m, n are coprime integers.
 $m = -1, n = 2$.
 $\gcd(1, 2) = 1$.
 $n > 0$.
 $m+n = -1+2 = 1$.
Wait, is it possible m, n are positive?
The value is negative. So m must be negative if we write m/n .
If the question implies $|val| = m/n$, it would say "absolute value" or similar.
Standard interpretation for m/n format with negative number is just
numerator/denominator.
One more check.
 $f(x+3)f(x-4) = -1$.
 $x = 0 \implies f(3)f(-4) = -1$.
 $x = 100 \implies f(1003)f(96) = -1$.
We don't need this.
We just need $f(-100) = -1/f(-73)$ (using $x-4 = -100 \implies x = -96 \implies f(-93)f(-100) = -1$, $f(-93) = -1/f(-93)$ equiv
 $3 \pmod{14}$, $f(3) = 5$).
Wait.
 $f(-93)f(-100) = -1$.
 $-93 = -7 \times 13 - 2 = -9 \times 1 - 2 \equiv -2 \pmod{14}$.
equiv $12 \pmod{14}$.
 $f(-93) = f(12) = -1/2$.
 $f(-1/2)f(-100) = -1 \implies f(-100) = 2$.
CONTRADICTION!
Let's re-evaluate this path.
 $f(-93)f(-100) = -1$.
We found $f(-100) = -1/2$.
So $f(-93)$ should be 2.
Let's check $f(-93)$ again.
 $-93 \pmod{14} = 1$.
 $93 = 14 \times 6 + 9$. ($14 \times 6 = 84$).
So $93 = 6 \times 14 + 9 \implies 93 \equiv 9 \pmod{14}$.
So $-93 \equiv -9 \pmod{14} \equiv 5 \pmod{14}$.
So $f(-93) = f(5) = 2$.
So $f(-93) = 2$.
Then $f(-93)f(-100) = 2 \times f(-100) = -1 \implies f(-100) = -1/2$.
Okay, my previous calculation of $-93 \pmod{14}$ was wrong in the "Wait" check.
 $14 \times 7 = 98$.
 $-93 = -98 + 5$. So $-93 \equiv 5 \pmod{14}$.
My previous check said $-93 \equiv 12 \pmod{14}$ based on wrong arithmetic.

tokens 6,189-6,954

$12 - 93 = -81$. Not divisible by 14.
Okay, the consistency check $f(-93)$ works.
 $f(-93) = f(5) = 2$.
 $f(-100) = -1/2$.
Okay, the result is robust.
Let's write down the steps clearly.
Derive periodicity $T = 14$ from $f(x+3)f(x-4) = -1$.
Substitute $x+4$ for x to get $f(x+7)f(x) = -1$.
Then $f(x+14) = f(x)$.
2. Reduce -100 modulo 14.
 $-100 = -8 \times 14 + 12 \equiv 12 \pmod{14}$.
So $f(-100) = f(12)$.
3. Use the relation $f(x+7)f(x) = -1$ to connect $f(12)$ to the interval $[0, 7)$.
Let $x = 5$. Then $x+7 = 12$.
 $f(12)f(5) = -1$.
 $f(12) = -1/f(5)$.
4. Calculate $f(5)$ using the definition $f(x) = \log_2(9-x)$.
 $f(5) = \log_2(9-5) = \log_2(4) = 2$.
5. Calculate result.
 $f(-100) = -1/2$.
6. Identify m, n .
 $m = -1, n = 2$.
7. Compute $m+n$.
 $m+n = 1$.
Is there any reason to doubt the format m/n ?
Sometimes fractions are required to be positive. But the value is clearly negative.
Unless the question implies $f(-100) = -\frac{\{m\}}{\{n\}}$?
"则 $f(-100)$ 的值为 $\frac{\{m\}}{\{n\}}$ 的格式, 请给出 $m+n$ 的值".
This literally means "the value of $f(-100)$ is in the format m/n ".
So if the value is -0.5 , it is $-1/2$.
So $m = -1, n = 2$.
If the question meant absolute value, it would say so.
Given it's a math problem context (likely Chinese Gaokao or similar competition style), m/n format with integers usually accepts negative numerators.
Also, usually m, n are coprime and $n > 0$.
Let's check if $f(x)$ could be complex? No, log base 2 usually implies real domain.
Domain $0 \leq x < 7$. $9-x$ is positive.
So $f(x)$ is real.
Everything holds in $\mathbb{R}$.
Final check of the question phrasing:
 $f(x+3) \cdot f(x-4) = -1$.
 $0 \leq x < 7$.
 $f(-100)$.
 m/n format.

Complete token-level view for the Figure 1 disagreement case | Part 5/5

Tokens 6,955-8,619 | Continuation of the same unabridged tokenizer-aligned response.

log r_t > 0 log r_t < 0

tokens 6,955-7,787

* m + n *. \n\nCalculation check : \n\$ 1 0 0 = 7 \ times 1 4 + 2 \$. \n\$ 1 0 0 \ equiv 2 \ p \ mod \ { 1 4 } \$. \n\$ - 1 0 0 \ equiv - 2 \ equiv 1 2 \ p \ mod \ { 1 4 } \$. \n\$ f (- 1 0 0) = f (1 2) \$. \n\$ f (1 2) = f (5 + 7) \$. \n\$ f (x + 7) f (x) = - 1 \$. \n\$ f (5 + 7) f (5) = - 1 \$. \n\$ f (5) = \ log _ 2 (9 - 5) = \ log _ 2 (4) = 2 \$. \n\$ f (1 2) = - 1 / 2 \$. \n\$ m = - 1 , n = 2 \$. \n\$ m + n = 1 \$. \n\nWhat if \$ n \$ can be negative ? \$ m = 1 , n = - 2 \$. \$ m + n = - 1 \$. \n\nStandard convention for fractions : denominator positive . \nI will stick with \$ n > 0 \$. \n\nStructure of the response : \n1 . Der ive periodic ity . \n2 . Reduce argument . \n3 . Use recurrence relation . \n4 . Evaluate using formula . \n5 . State m , n , and sum . \n\nWait , is there any possibility that \$ f (x + 3) f (x - 4) = - 1 \$ implies something else ? \n\$ f (x + 7) = - 1 / f (x) \$. \n\$ f (x) \ neq 0 \$. \nIf \$ f (5) = 2 \$, then \$ f (1 2) = - 0 . 5 \$. \nThen \$ f (1 9) = - 1 / (- 0 . 5) = 2 \$. \nThen \$ f (1 9) = f (5 + 1 4) \$, consistent . \n\$ f (2 6) = f (5 + 2 1) = f (- 1 0 0 + ?) \$. \n\$ - 1 0 0 = 1 9 - 1 1 9 = 2 + 1 7 - \dots \$. \n\$ f (- 1 0 0) = f (1 2) = - 0 . 5 \$. \n\$ f (1 2) = f (1 2 - 1 4) = f (- 2) \$. \n\$ f (- 2) = - 1 / f (5) = - 1 / 2 \$. \n\$ f (1 2) = - 1 / f (5) = - 1 / 2 \$. \nCons istent . \n\nIs there any ambiguity with \$ x = 5 \$? \n\$ 0 \ le 5 < 7 \$. Yes . \nIs there any boundary issue ? No . \nIs it possible to use \$ f (x - 4) \$ relation directly on \$ f (- 1 0 0) \$? \n\$ f (x - 4) = f (- 1 0 0) \ \implies x - 4 = - 1 0 0 \ \implies x = - 9 6 \$. \n\$ f (- 9 6 + 3) f (- 1 0 0) = - 1 \ \implies f (- 9 3) f (- 1 0 0) = - 1 \$. \n\$ - 9 3 \ equiv 5 \ p \ mod \ { 1 4 } \$. \n\$ f (- 9 3) = f (5) = 2 \$. \n\$ 2 \ cdot f (- 1 0 0) = - 1 \ \implies f (- 1 0 0) = - 1 / 2 \$. \n\nThis is a faster route to verify without explicitly stating period 1 4 first , just linking via mod 1 4 or just successive 7 -sh iffs . \n\nWait , I used mod 1 4 to get \$ f (- 9 3) = f (5) \$. \n\$ - 9 3 = - 1 4 \ times 7 - 5 \$. No . \n

tokens 7,788-8,619

\$ 1 4 \ times (- 7) = - 9 8 \$. \$ - 9 3 = - 9 8 + 5 \$. Correct . \n\nSo \$ f (- 9 3) = f (5) \$. \nThis confirms the result without explicitly calculating \$ - 1 0 0 \ p \ mod \ { 1 4 } \$. \n\$ - 9 3 \ equiv 5 \$. \n\$ 5 \ in [0 , 7) \$. \n\$ f (5) = 2 \$. \n\$ 2 \ cdot f (- 1 0 0) = - 1 \ \implies f (- 1 0 0) = - 0 . 5 \$. \n\nBoth methods align perfectly . \nFinal check on \$ m + n \$. \n\$ m = - 1 , n = 2 \ \implies m + n = 1 \$. \n\nThe question asks for * m + n *. \nIs it possible they want \$ | m | + n \$? No , just * m + n *. \nIs it possible \$ f (- 1 0 0) \$ is an integer ? No . \n\nOkay , ready to write . \n\n</think> \n\n由已知条件 \$ f (x + 3) \ \cdot \ f (x - 4) = - 1 \$, \n令 \$ x = \$ 替换为 \$ x + 4 \$, 则有 : \n\$\$ f ((x + 4) + 3) \ \cdot \ f ((x + 4) - 4) = - 1 \$\$ \n即 \n\$\$ f (x + 7) \ \cdot \ f (x) = - 1 \$\$. \n\n由此可得 \n\$\$ f (x + 7) = - \ \frac { 1 } { f (x) } \$\$. \n再次应用 此关系, 将 \$ x = \$ 替换为 \$ x + 7 \$: \n\$\$ f (x + 1 4) = f ((x + 7) + 7) = - \ \frac { 1 } { f (x + 7) } = - \ \frac { 1 } { f (x) \cdot (- 1 / f (x)) } = f (x) \$\$. \n说明函数 \$ f (x) \$ 是 周期函数, 且 周期 \$ T = 1 4 \$. \n\n我们需要求 \$ f (- 1 0 0) \$ 的值 . \n\n利用周期性, 我们可以将 自变量 \$ - 1 0 0 \$ 转化为一个在已知定义域 \$ [0 , 7) \$ 附近的值 . \n\n计算 \$ - 1 0 0 \$ 关于 \$ 1 4 \$ 的同余关系 : \n\$\$ - 1 0 0 = 1 4 \ times (- 8) + 1 2 \$\$. \n即 \$ - 1 0 0 \equiv 1 2 \ p \ mod \ { 1 4 } \$. \n\n所以 \n\$\$ f (- 1 0 0) = f (1 2) \$\$. \n\n接下来利用 \$ f (x + 7) \ \cdot \ f (x) = - 1 \$ 将 \$ f (1 2) \$ 转化到区间 \$ [0 , 7) \$. \n\n因为 \$ 1 2 = 5 + 7 \$, 且 \$ 5 \ in [0 , 7) \$, 所以 : \n\$\$ f (1 2) \ \cdot \ f (5) = - 1 \ \implies f (1 2) = - \ \frac { 1 } { f (5) } \$\$. \n\n又因为当 \$ 0 \ le q \ slant x < 7 \$ 时, \$ f (x) = \ log _ 2 (9 - x) \$, \n\n所以 \n\$\$ f (5) = \ log _ 2 (9 - 5) = \ log _ 2 (4) = 2 \$\$. \n代入上式, 得 : \n\$\$ f (1 2) = - \ \frac { 1 } { f (2) } \$\$. \n\n因此 \n\$\$ f (- 1 0 0) = - \ \frac { 1 } { f (2) } \$\$. \n\n题目要求将结果表示为 \$ \frac { m } { n } \$ 的形式, 并求 \$ m + n \$ 的值 . \n\n这里 \$ m = - 1 \$, \$ n = 2 \$ (通常约定分母 \$ n \$ 为正整数). \n\n则 \$ m + n = - 1 + 2 = 1 \$. \n\n\boxed{1}